

An Introduction to Open Source Platforms *for* Developing Mobile Applications

The superabundance of mobile devices and operating systems makes mobile application development a challenging task. The ideal is to develop applications that work on all devices by using a single code base and yet give the apps a native feel. Let us briefly look at a few open source mobile app development platforms that help developers do so.

As of the first quarter of 2018, we could choose from about 3.8 million apps on Google Play Store and 2 million on Apple's App Store (as per Statista2018). By 2020, mobile apps are expected to generate around US\$ 189 billion in revenue via different app stores and from in-app advertising. All this illustrates the growing culture of mobile app development.

The development of any mobile app has its roots in traditional software development. But, the end result needed is software that makes use of the unique features and hardware of different mobile devices. Mobile applications must be designed to give optimal performance within the constraints of the mobile phones' hardware. For instance, a mobile gaming application will have limited graphical elements as mobile devices have limited graphics processors. Solving the performance issue on any given device mainly depends on developing apps natively on that specific device. So code needs to be designed specifically for the given hardware on a particular device. This is quite easy in the case of iOS devices, since developers only require versions of the application for the iPhone and iPad in order to achieve universal usability. However, all the other smartphones or tablets run on different hardware systems or on varying versions of the same operating system.

Mobile app development covers a set of different procedures and processes. As we know, these applications are more like desktop software, and are designed using a wide range of programming languages and software development frameworks. These mobile apps can be pre-installed on various mobile phones or can be delivered as



Web applications using client-side and server-side processing to create an 'application-like' experience within a browser as well. Mobile app developers need to consider the following parameters while developing an application:

1. The operating system supported by the mobile phone
2. The different hardware specifications like the processor speed, RAM, etc, of the mobile phones on which the app needs to run
3. The configuration of the mobile phone
4. The wide range of screen sizes of the different mobile phones
5. The programming language used for developing mobile apps. It could be objective-C, Java, etc
6. The platforms that will be used for development, deployment and the management of different mobile apps

Mobile app development services

Any mobile app development process requires access to the software development kits (SDKs). SDKs provide an environment with which programmers can design and test the code in a simulated mobile environment. Developing a mobile app does not require full usage of these kits; for instance, various mobile games can be developed entirely using Unity and then the Android SDK can be used to simply ensure its delivery on mobile devices. Developing applications for iOS mobile phones requires a paid iOS developer licence, whereas the Android SDK is available free of cost to users.